



Intensity duration and frequency of Heat wave in different phases of MJO over India

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ARTICLE INFO

Keywords:

Intensity
Duration
Frequency
MJO
Heat wave
Extreme weather event

ABSTRACT

The Madden-Julian Oscillation (MJO) is an important tropical atmospheric phenomenon that plays a significant role in modulating weather patterns across India. Heat Waves (HW) are extreme weather events that have severe impacts on human health, agriculture, and various socio-economic sectors. This research paper aims to investigate the dynamical relationship between MJO phases, as measured by the Real-time Multivariate MJO (RMM) index, and the occurrence of HW in India, as defined by the Indian Meteorological Department. This paper demonstrated the relationship among three aspects, i.e., intensity, duration, frequency of HW during the different MJO phases. The differential dynamics associated for about two-fold rise in the HW occurrence in the South East, Eastern and North West India in the MJO active phase. Inactive phase witnessed enhancement (suppression) of HW during wet (dry) over the regions. The novel finding of this paper is that, the weaker the MJO, the higher probability of HW occurrence in respective regions depending on the position of the MJO convection center. The maximum duration is observed to be significantly decreased by about 2 days in the active MJO phase over NW and north central parts of the country. It is also inferred from this study that there is a significantly decrease of 0.5 °C in HW intensity during the active wet phase of MJO over the NW India. To explain the linkage of HW with MJO, geopotential height (GPH), wind circulation, velocity potential (VP), sea surface temperature (SST) and outgoing longwave radiation (OLR) composite anomalies have been computed and used in the analysis. It is observed that all these parameters have a direct impact on the evolution, strength and spread of the HW in India. This dynamical study mechanism can be prescribed in the numerical weather prediction models for the improved prediction of such extreme weather events during summer for proactive disaster management.

1. Introduction

In the present climate change scenario, anthropogenic forcing has the major contribution in shifting the temperature towards the higher side (IPCC, 2021). The intensity-duration-frequency of extreme weather events (EWE) over the whole globe is increasing (Meehl and Tebaldi, 2004; Coumou and Rahmstorf, 2012; Rohini et al., 2016). Heat Waves (HW) are characterized by prolonged periods of excessively high temperatures, often accompanied by high humidity, which can have devastating effects on human health, agriculture, and the economy (Khan et al., 2019). With the increasing maximum and minimum temperatures over the whole globe, the occurrence of HW is projected to be more intense, longer and frequent (Perkins et al., 2012; Perkins and Alexander, 2013; Panda et al., 2017). India experiences frequent HW

during the summer months extending from April till June every year. The deadliest HW (1998, 2010, 2012, 2015) has occurred in India in the last three decades and has taken up millions of lives in the past 30 years (Ratnam et al., 2016; Mazdiyasni et al., 2017; Ray et al., 2021). Rising maximum and minimum temperatures pose serious health risks to people of India in future as there is a worrisome trend with projected catastrophic HW in the northern, northwest, central, and eastern coastal regions from April to June (Rohini et al., 2016; Neethu et al., 2020; Singh et al., 2021; Devi et al., 2023). The average frequency of HW events during April–June months is 1–2 and lasts for 5–7 days over the north western and southeastern parts of India (Rohini et al., 2016; Dodla and Satyanarayana, 2021). The HW frequency, total duration and maximum duration has significantly increased by 1 event/20 year, 2 days/decade and 1.5 days/decade over north western region in India

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<https://doi.org/10.1016/j.atmosres.2024.107250>

Received 31 August 2023; Received in revised form 12 January 2024; Accepted 13 January 2024

Available online 14 January 2024

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(Rohini et al., 2016). India experiences two main types of HW: dry and moist. The dry HW, prevalent in the northwestern region, are marked by stable conditions, clear skies, and intense solar radiation due to a blocking pattern over the North Atlantic Ocean (Ratnam et al., 2016; Goyal et al., 2023). On the other hand, moist HW in the southeastern coastal region, associated with the Matsuno-Gill response to cooling in Pacific Ocean brings high temperatures and humidity (Goyal et al., 2023).

The HW occurrence has been linked with hot winds transport from Pakistan into India and blockage of hot air due to anticyclonic motion as the ridge of the subtropical jet stream falls over (Ratnam et al., 2016; Singh and Kumar, 2018; Dubey et al., 2021). It is shown that the eastward shift of upper-level air is linked with clear skies, resulting in an increase in accumulated heat and land surface temperature (Ratnam et al., 2016). The presence of hot air mass in the arid and desert region over north western region leads to the decreased convective clouds and increased HW in the region (Goyal et al., 2023). The HW occurrence over India mainly driven by the advection and subsidence of warm air and these are the dynamical processes which are influenced by the synoptic scale processes (Dodla and Satyanarayana, 2021).

The Interannual and Intraseasonal oscillation in the tropical ocean on a large scale affects the HW occurrence over India (De and Mukhopadhyay, 1998; Revadekar et al., 2009). The equatorial Rossby waves has a significant role in exacerbating the HW condition over India. The presence of the positive phase of the equatorial Rossby wave leads to double the probability of HW events of longer duration (3–7) days and higher intensity (above 44 °C) over India (Zore et al., 2021; Lekshmi and Chattopadhyay, 2022). The HW over India also has been influenced by the anomalous sea surface temperature (SST) in the Pacific Ocean (Goyal et al., 2023). The El Niño Southern Oscillation (ENSO) is a stationary event, lasts for >8 months to 1.5 years and modulates the characteristics of HW occurrence in its positive (negative) phase i.e., El Niño (La Niña). It is observed that during the El Niño years, HW is longer as compared to neutral years as the maximum temperature rises by about 1 °C (Murari et al., 2016). It is quantified that HW days are increasing by 0.2 to 0.5 days per year (decreasing by 0.1 to 0.2 days per year) in El Niño+1 (El Niño –1 and El Niño) years (Satyanarayana and Dodla, 2020; Naveena et al., 2021). As the tropical Indian Ocean continues to warm, coupled with an increasing occurrence of El Niño events in the future, India may experience a heightened frequency and prolonged duration of HW episodes (Rohini et al., 2016). The impact of ENSO on HW has been highlighted and for 53 years of Madden Jullian Oscillation (MJO) concept evolution, its relationship with HW is still under cover. The non-stationary MJO phenomenon is an eastward movement of enhanced convection and subsiding air in a period of 30 to 60 days over the whole globe (Julian and Madden, 1981; Lau and Chan, 1986; Zore et al., 2021). Basically, the phase of MJO locates the enhanced convection across the whole globe during the 30–60 days of MJO lifecycle (Wheeler and Hendon, 2004; Xu and Rutledge, 2014). The enhanced convection leads to the adiabatic cooling and heavy precipitation, so considered as the wet phase, while the suppressed convection or subsidence of air leads to adiabatic heating of the surface and also the humidity will decrease, so considered as the dry phase (Dubey et al., 2021; Almazroui, 2023; Lenka et al., 2023). The MJO may interfere with the upper air circulation (jet streams) and when the MJO wave falls over the jet stream, the jet typically bends northward over the western part of the country and southward to the east (Wanying and Tziperman, 2018; Dubey et al., 2021). MJO plays an important role in the upper tropospheric circulation affecting the hot air imported from the temperate zone towards the equatorial region. That mechanism leads to unseasonably warm temperatures in the west and cool temperatures in the east (Matsueda and Takaya, 2015; Wanying and Tziperman, 2018). The HW intensification due to the enhanced tropical convection is observed over Australian region (Parker et al., 2014). The active phases of MJO in the Indo-Pacific region leads to the changes in Rossby wave train patterns at upper level which has been related to the HW events (Song and Wu, 2019; Hsu et al.,

2020; Rodrigues et al., 2019). The direct relationship of MJO with monsoon onset and seasonal rainfall has been well understood in various studies (Subbaramayya and Suryarao, 1976; Ramachandra et al., 1988; Wheeler and Hendon, 2004; Mishra et al., 2017; Taraphdar et al., 2018; Baburaj et al., 2022; Lenka et al., 2023), but the localized linkage of MJO phases with HW is yet to be explored. The MJO can influence the initiation and propagation of Rossby wave in the upper troposphere over subtropical region, which in presence of subtropical westerly jet stream can modulate the HW characteristics over India (Song and Wu, 2019; Barpanda et al., 2023). The tropical convection and suppression significantly affect the HW occurrence over India (Zore et al., 2021). During the positive (negative) phase of active MJO, southeastern (SE) India faces more (less) frequent HW events. The opposite effect has been observed in the north western (NW) region in India (Zore et al., 2021). However, the influence of different phases of MJO on HW characteristics in India has not been explored extensively. The primary objective of this work is to investigate the effect of eight MJO phases, as measured by the Real-time Multivariate MJO (RMM) index, on the occurrence and characteristics of HW in India. The main objectives of this study are given below:

- Analyzing the relationship between MJO phases and the frequency of HW at a localized scale in different regions of India.
- Assessing the impact of MJO phases on the intensity and duration (maximum and average) of HW in India.
- Identifying potential physical mechanisms and teleconnections that link MJO phases with the occurrence of HW in India.

Understanding the relationship between MJO phases and Indian HW is of significant importance for agriculture, livestock, and various socio-economic sectors. By establishing a relationship between MJO phases and HW occurrence, can enhance early warning systems and enable authorities to take timely measures to mitigate the impacts of the heat disasters (Lee and Grotjahn, 2016). With the ongoing changes in global climate patterns, it is crucial to understand how natural climate modes like the MJO interact with the regional HW dynamics (Hsu et al., 2020). This knowledge can help in developing effective adaptation and mitigation strategies to minimize the adverse effects of HW disasters in a changing climate. By addressing these objectives and highlighting the importance of the study, this research aims to contribute to the scientific understanding of the complex dynamical interactions between the large-scale MJO and regional HW characteristics in India.

2. Materials and methods

The daily maximum temperature (DMT) gridded dataset of $1^\circ \times 1^\circ$ resolution from the Indian Meteorological Department for period of 62 years, i.e., 1961–2022 (Srivastava et al., 2009) are used in the present analysis. To carry out the spatio-temporal analysis of the HW event, the DMT data has been bilinearly interpolated to a higher resolution $0.25^\circ \times 0.25^\circ$. Considering the frequency and intensity of such disaster events in the summer season, all the analysis are carried out for the period 1st April - 30th June every year. The HW criterion as defined by the Indian Meteorological Department (Pai et al., 2014; Singh and Kumar, 2018; Malasala et al., 2020; Devi et al., 2023) has been adopted in this analysis to define the HW at each grid point in the country and this is summarized in Table 1. The normal temperature for HW dates is being selected based on the daily climatology (Norm) for the period 1961–1990 in line with the earlier study (Rohini et al., 2016). The daily temperature anomaly (Anom) has been computed using Norm. All the composite figures have been tested for the 90% level of significance using two-tailed student's *t*-test.

In this study, the whole of India has been divided into three physiological regions namely 1. plains (elevation <600 m), 2. hilly (elevation >600 m), and 3. coastal region. The areas under 100 km from the coastline of the Indian Ocean have been marked as the coastal region

Table 1

HW criterion used to define the HW over India.

Criterion	Plains	Coastal	Hilly
(a) Based on the DMT anomaly	If Norm >40 °C & Anom >4 °C If Norm <40 °C & Anom >5 °C	If Anom >4 °C	If Norm >30 °C & Anom >4 °C If Norm <30 °C & Anom >5 °C
(b) HW should not be considered until	If DMT < 40 °C	If DMT < 37 °C	If DMT < 30 °C
(c) HW may be declared Irrespective of Norm.	If DMT ≥ 45 °C	–	–

(Srivastava et al., 2022). The elevation data has been used from GTOPO30, 1 km resolution data from the U.S. geological survey digital elevation model (USGS EROS Archive, GTOPO30, 2022). This elevation data has been further extrapolated to 25 km for better analysis.

The HW condition as shown in Table 1, if satisfied for consecutive 2 or more days at one grid point has been noted as a HW event. A total of 137 HW events, spanning 910 HW days, were identified over India during a 44-year study period from 1979 to 2022. The grid wise computation of the HW parameters is considered using the following equations

$$\text{Intensity} = \text{DMT} - \text{Norm}$$

$$\text{AverageDuration} = \frac{\text{TotalnumberofHWdays}}{\text{TotalnumberofHWEvents}}$$

$$\text{MaximumDuration} = \text{DurationofLongestHWEvent(days)}$$

The intensity of a HW refers to how hot and prolonged a period of excessively high temperatures is observed in a particular region. The intensity of HW has been measured by the composite anomaly of DMT during the identified HW events (Zore et al., 2021; NOAA, 2022). The length or period of a HW event is denoted as the HW duration in days, computed for all HW events during the 44-year study period. To calculate the average maximum duration of HW, the longest HW event for each year is identified and averaged over the years.

The strength and progression of the MJO through eight different areas along the equator around the world are monitored by the Bureau of Meteorology (BoM). The Realtime Multivariate MJO (RMM) index has been used by the Australian Bureau of Meteorology (<http://www.bom.gov.au/climate/mjo/>). The RMM index provides information on the strength and phase of the MJO, which is essential for analyzing its relationship with HW. It is derived from the combined signals of zonal wind and Outgoing Longwave Radiation (OLR) anomalies in the tropical region (Wheeler and Hendon, 2004). The RMM index consists of two components: RMM1 and RMM2, representing the eastward and northward propagation of the MJO, respectively. The two geometric axes of the phase-space diagram are used to quantify values of the principal components, capturing the strength and direction of the MJO's progression over time. The RMM index is being used to determine the phase of the MJO during specific periods.

The whole analysis has been performed for the active (RMM index ≥ 1) and inactive (RMM index < 1) phases of MJO. These two active and inactive phases have been further divided into wet and dry phases for a better representation of the MJO dynamics (Lee and Grothjahn, 2016; Lenka et al., 2023). To enhance the robustness of the analysis, further investigations were conducted within four subcategories. This approach was taken due to the limited number of HW days in individual phases during HW events. The Indian monsoon onset over Kerala generally occurs in the first week of June (Lenka et al., 2023). So, the different clusters in dry and wet phases of MJO in pre-monsoon and monsoon season is due to the June month HW. The previous studies have reported the different clusters of MJO phases for June – September (Mishra et al., 2017; Taraphdar et al., 2018; Lenka et al., 2023) based on the positive (negative) anomaly of rainfall, Indian summer monsoon onset over

Kerala. The phase clusters (1–4) and (5–8) have been made based on the significant change in the probability of HW in 8 phases of MJO and phase cluster (1–4) shows the enhancement of HW occurrence probability over India (Fig. 1a-i). The wet (negative OLR anomaly) and dry phase (positive OLR anomaly) have been chosen based on the observed outgoing longwave radiation anomaly for April – May (fig. S1). The HW over India occurs in the pre-monsoon summer season (April–June) and the maximum occurrence is in May (Naveena et al., 2021). The OLR anomaly distribution in June (Fig. S2) shows the shift because of the monsoon onset and progress thereof in India (Lenka et al., 2023). So, the different clusters of MJO phases in dry and wet phases in pre monsoon (April–June) and monsoon (June–September) seasons are possible. To remove the impact of the monsoon onset rainfall effect, the OLR anomaly for April–May has been considered to declare wet and dry phase. With the above classification the number of MJO days as well as HW days as observed in different MJO phase combinations (A1234, A5678, IA1234, IA5678) are counted (Table 2) and are used for further analysis.

The probability of HW is computed in terms of phase and climatological value as follows.

$$P(\text{Phase}) = \frac{\text{HW}_{\text{DaysMJO}}}{\text{TotalDaysMJO}} \text{ and } P(\text{Clim}) = \frac{\text{TotalHW}_{\text{Days}}}{\text{SummerDays}(91 \times 44.)}$$

The total number of days in the summer season considered is (1st April to 30th June) i.e., 91 for 44 years of study. Then the change in probability of HW occurrence, denoted as $P(\text{change})$ is computed by using the probability of HW occurrence in each phase of MJO $P(\text{Phase})$ and the climatological probability of HW i.e., $P(\text{Clim})$.

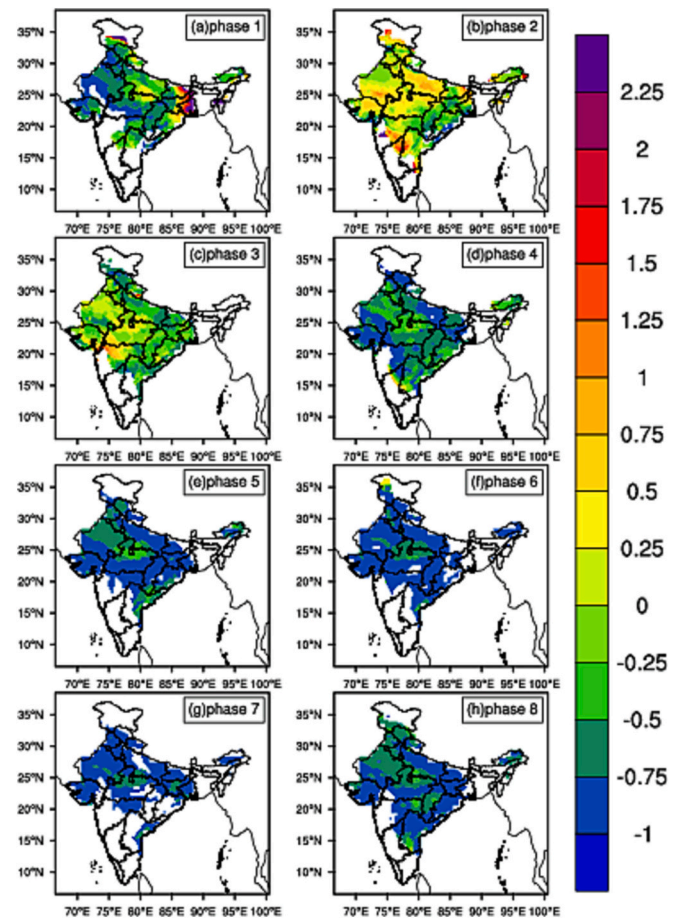


Fig. 1. (a) – (h) Spatial distribution of change in probability of HW occurrence during all phases of MJO (phase 1 to 8) over the Indian region for the period 1979–2022. The areas shaded indicate a significance level of 90%.

Table 2

Distribution of HW days in different phase of MJO during summer season.

Number of Days	MJO Phase Type	Active	Inactive
Total MJO days	Dry (5678)	1166	735
	Wet (1234)	1264	748
Total MJO days during HW at any Grid Point	Dry (5678)	292	151
	Wet (1234)	249	218

$$P(\text{Change}) = \frac{P(\text{Phase}) - P(\text{Clim})}{P(\text{Clim})}$$

The daily mean of OLR, geopotential height (GPH), zonal and meridional wind at $2.5^\circ \times 2.5^\circ$ resolution has been taken from National Centre for Environment Prediction (NCEP) reanalysis 2 dataset (Kanamitsu et al., 2002). National Oceanic and Atmospheric Administration (NOAA) interpolated SST (Huang et al., 2021) and interpolated OLR (Liebmann and Smith, 1996) data is used to understand the dynamics better. To understand the role of convergence and divergence at 850 hPa and 200 hPa, velocity potential (VP) and OLR anomalies during the HW days in the combined MJO Phases i.e. A1234, A5678, IA1234, IA5678 during HW are computed. This study focused on the composite analysis of VP, SST, and OLR anomalies during HW days to assess MJO and linked atmospheric/oceanic conditions in the Indian HW scenario. The anomalies have been computed concerning the long-term daily mean for the period 1979–2022. This analysis will help determine if certain MJO phases are associated with a higher or lower frequency of HW in Indian land mass. The duration and intensity of HW during different MJO phases have been also examined at a regional scale.

3. Results

3.1. Relationship between MJO Phases and HW occurrence

This analysis is mainly focused on determining whether specific MJO phases and strength are associated with an increased or decreased likelihood of HW events. To investigate the dynamical relationship between MJO and HW occurrence, the significant change in probability of HW occurrence is being analyzed during all 8 MJO phases irrespective of MJO strength (Magnitude of RMM index). Climatologically during MJO phase 1, the occurrence of HW is almost 2.25 times more likely than normally over the states and regions like West Bengal (WB), Bihar, Jammu & Kashmir(J&K), Mizoram and Tripura. The HW occurrence probability is 0.5 times more than normal over parts of Uttar Pradesh (UP), Odisha and Arunachal Pradesh (ARP) in phase 1 as depicted in Fig. 1a. Similarly, analysis indicates that the HW occurrence is 0.5–1.25 times more likely over the parts of UP, Madhya Pradesh (MP), Rajasthan (RJ) and Bihar in MJO phase 2 and a maximum of 1.5–2.25 times increment is observed over parts of Maharashtra (MH), Karnataka, Telangana and Andhra Pradesh (AP). The MJO phase 2 shows a higher probability of HW in most parts of India except eastern and western coastal region (Fig. 1b). The MJO phase 3 show about 0.25–0.5 times more likely chances of HW occurrence in parts of MH, MP, Himachal Pradesh (HP) and Gujrat state (Fig. 1c). Simultaneously, MJO phase 3 shows up to 0.5 to 0.75 times reduction in HW occurrence over the eastern, northern, central and south-eastern parts of India. The MJO phase 4 (Fig. 1d) shows reduced chances of HW occurrence by –0.25 to –1 times than normal over all the parts of India except north Karnataka and Telangana. Whereas phases 5, 6, 7, 8 of MJO show the complete suppression of HW occurrence by –0.5 to –1 over the whole India as explained in Figs. 1e–1h. Among all the phases of MJO, the 7th phase (which generally falls in June) witnessed the least HW during the 44-year analysis and this could be due to the absence of any convective activity in India due to the onset phase of the monsoon.

The all-India area average analysis shows that HW occurrence is significantly less likely in all the MJO phases except phase 2 (Fig. 2),

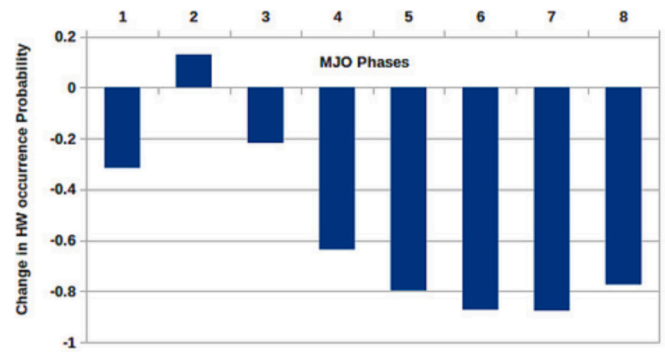


Fig. 2. Temporal distribution of significant change in probability of HW occurrence during MJO active phases over the India as a whole during the period of 44 years (1979–2022).

hence the temporal distribution suggests relief in HW occurrence over India during all the phases except phase 2. The MJO phase-wise quantification of change in HW occurrence probability suggests that HW events are more (less) likely during phases 1–4 (5–8) as shown in Fig. 2. This study marks the first comprehensive analysis and reporting of the specific influence of RMM index-based MJO phases and strength on the occurrence, intensity, and duration of HW in India. The change in HW occurrence probability in MJO active dry phase is negative over most of the parts of the country i.e. suppressed by up to –0.8 times than climatological values (Fig. 3a). The HW probability of occurrence in the active dry phase is significantly enhanced by 0.8 times in the eastern India (Bihar, Jharkhand, ARP), Himalayan states (Himachal Pradesh, Uttarakhand) and 2 to 2.4 times in the SE state of AP (Fig. 3a). The significantly increased probability of HW occurrence has been observed in the active wet phase over north-eastern (NE), WB, Bihar, Jharkhand, western, NW and central parts of India (Fig. 3b). The highest increment (2-fold) in HW occurrence probability has been observed in the active wet phase over the parts of MH and NE states. There is total suppression of HW occurrence observed over Chhattisgarh, Jharkhand, Odisha and SE India in the active wet phase whereas enhancement is seen in the active dry phase over the same region (Fig. 3b). The HW have been completely suppressed (–1) in the inactive dry phase over most of the parts of India except Chhattisgarh, NW, northern and NE (Fig. 3c).

With the weaken MJO, HW occurrence probability has been significantly decreased in inactive dry phase as compared to active dry phase over the same regions. The HW occurrence probability in the inactive wet phase have been significantly increased by double over whole of India. The maximum enhanced of HW occurrence probability (2.4 times than normal) have been observed over western, central, WB, Uttarakhand, Jharkhand, north Karnataka, Telangana, AP and MH parts in India. The probability of HW occurrence has been observed to increase by nearly 2.8 times over Chhattisgarh, Odisha, and AP (Fig. 3d). Hence MJO active wet and dry phases have regional effects on eastern coastal and rest of the parts of country. The influence of the MJO strength has been notably observed, leading to a significant increase (decrease) in the probability of HW occurrence during inactive (active) events over the corresponding regions in India (Fig. 3). The temporal distribution of averaged significant change in HW occurrence probability over the whole India shows that the inactive wet phase leads to doubling the climatological probability and the dry phases leads to complete suppression (Fig. 4). Overall, the wet (dry) phase leads to the significant enhancement (suppression) of the HW occurrence (Fig. 4).

3.2. Duration of HW in different MJO Phases

The composite climatological maximum duration of HW events during active wet, active dry, inactive wet and inactive dry have been computed and presented in Fig. 5. The longest HW episode of about

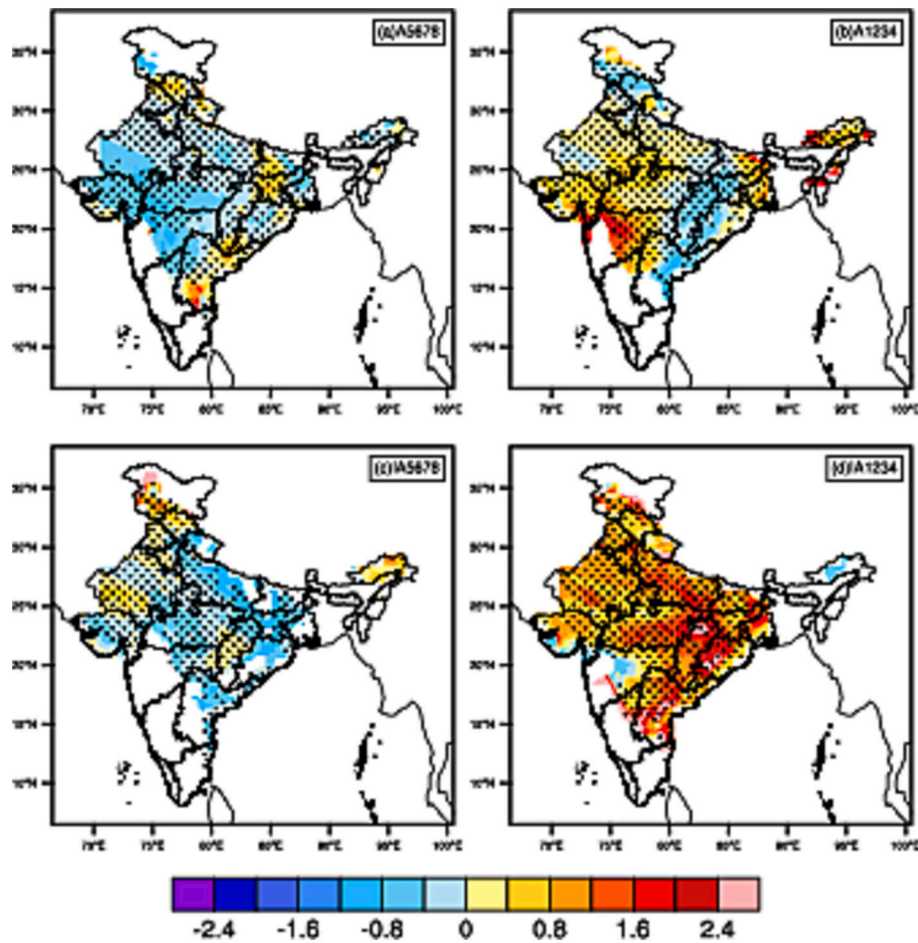


Fig. 3. Spatial distribution of change in probability of HW occurrence during different combination of MJO phases i.e. (a) active 5678 (b) active 1234 (c) inactive 5678 (d) inactive 1234 over the Indian region during the period of 44 years (1979–2022). The areas marked with dot signify a 90% level of significance.

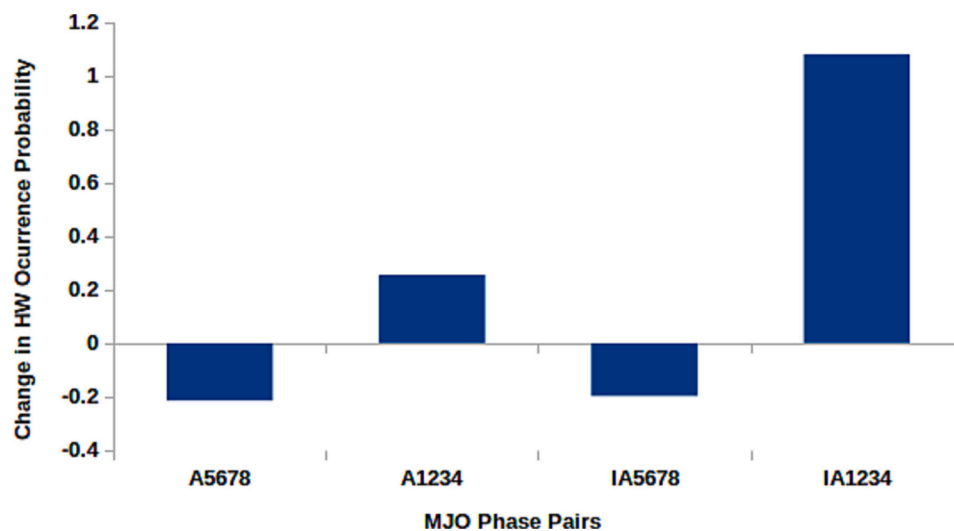


Fig. 4. Temporal distribution of significant change in probability of HW occurrence during MJO phase combinations over India as a whole during the period of 44 years (1979–2022).

12–16 days has been observed over the Vidarbha region in MH and Ahmedabad, Vadodara in Gujarat state during all the phases of MJO. It is observed that the recorded maximum duration of HW is >10 (6) days in the inactive (active) phase all over India except the peripheral regions like Indo-Gangetic plains and interior southern plain (Fig. 5 a-d). The

phase of MJO has a localized effect on northern MP, NW and north central parts as, the maximum duration is higher by 2 days in the wet phase as compared to the dry phase (Fig. 5 a, b). The maximum duration has been found significant (90% level) during all the phase clusters except the active wet phase. During the inactive phase (weaker MJO),

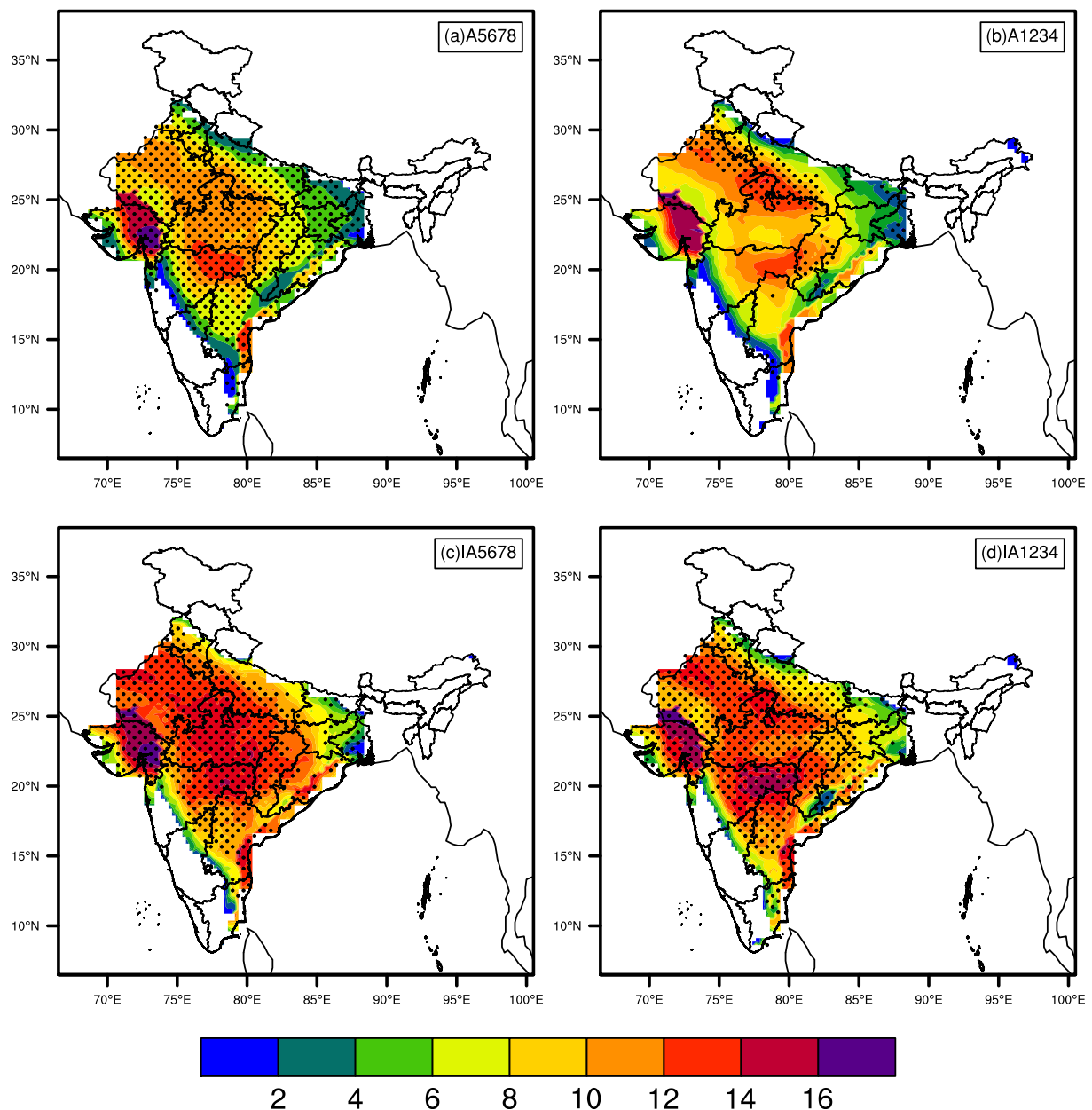


Fig. 5. Spatial distribution of composite climatology of maximum duration of HW events in days in MJO [(a) active 5678 (b) active1234 (c) inactive 5678 (d) inactive1234] phases during the identified HW events over Indian region. The units of the shading considered is days/year. The areas marked with dot signify a 90% level of significance.

there is an observed increase in the maximum duration by approximately 2 days in the northern region of MP, west coastal and central region of India.

3.3. HW Intensity in Different MJO Phases

The impact of the MJO phase and strength on the HW intensity has been quantified in terms of change in DMT during HW events over India. The composite DMT anomaly (intensity) compared to the climatological mean (1961–1990) varies from 3.75 °C to 6.75 °C during the HW events (Fig. 6). Higher intensity has been observed over the northern, NE and eastern parts of the country. The coastal region and associated areas face the humid HW where humidity plays an important role in the HW intensification and persistence. The eastern states (Chhattisgarh, Jharkhand, Bihar & Odisha) face HW intensity of 5.25 °C to 6.75 °C in the active wet, active dry and inactive wet phases of MJO (Fig. 6 a, b, d).

In the inactive dry phase, the lowest anomaly 3.5 °C to 4.5 °C has been observed over the central states of India (Fig. 6 c). With the enhanced convection in active phases, significant decline (0.5 °C) in Intensity over NW regions have been observed as compared to inactive phases. The HW intensity has been influenced by the strength of MJO whereas phase has an impact on specific locations (i.e., AP, Northern MP, Vidarbha). The HW intensity is observed to be reduced by 0.5 °C with the strength of MJO and is higher during the wet phases.

3.4. Analysis of Atmospheric Environment during HW in different MJO Phases

This section provides insights into the mechanisms by which the MJO influences HW dynamics in different regions of India. The major parameters like OLR, GPH and VP along with the oceanic parameter SST are considered to establish their role in modulating the MJO phases and

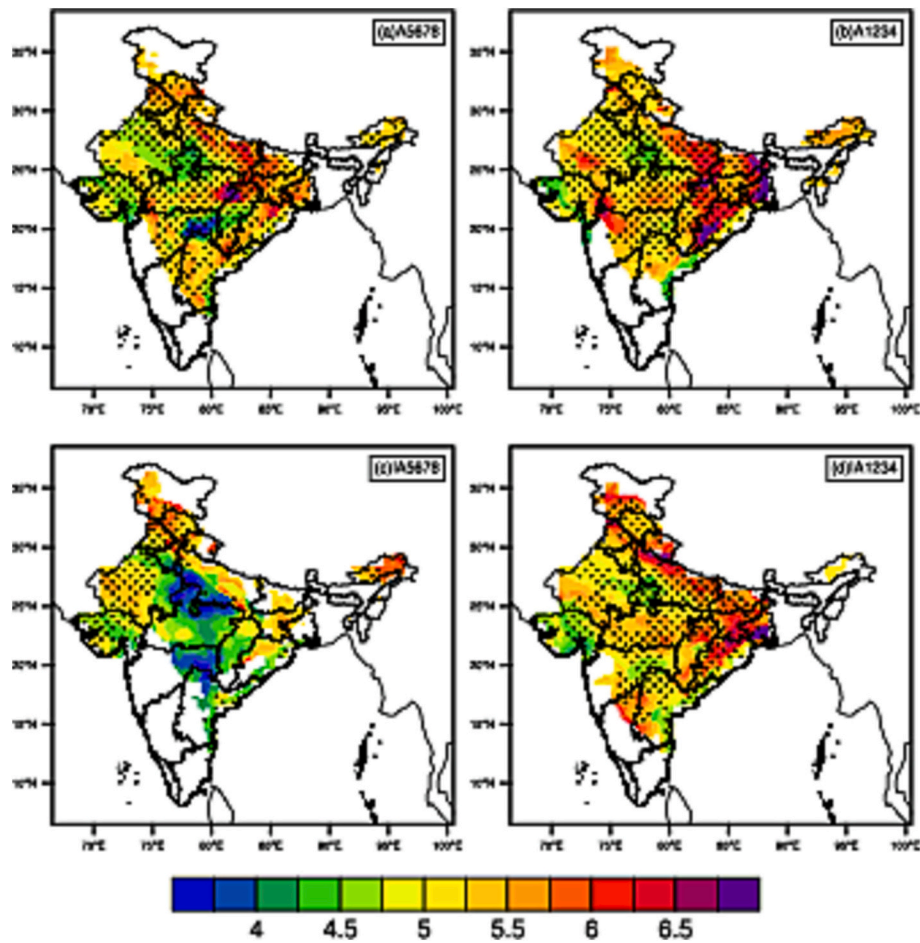


Fig. 6. Spatial distribution of composite anomaly of DMT in MJO [(a) active 5678 (b) active1234 (c) inactive 5678 (d) inactive1234] phases during the identified HW events over Indian region. The unit of the shading used is $^{\circ}\text{C}/\text{day}$. The areas marked with dot signify a 90% level of significance.

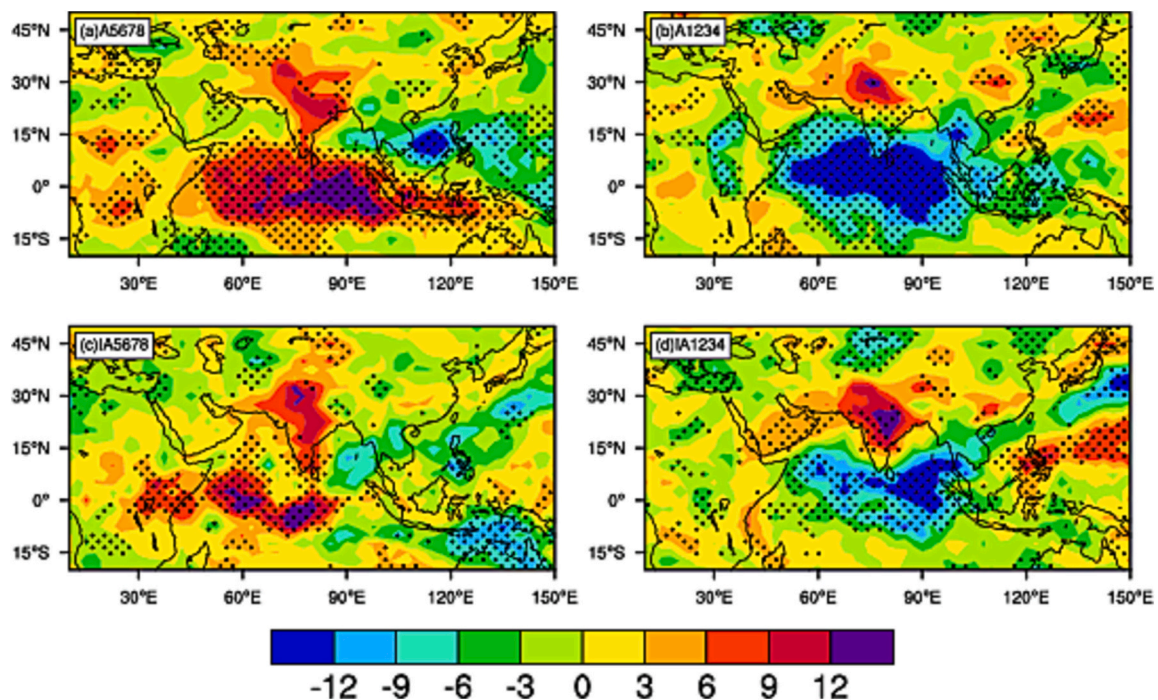


Fig. 7. Spatial distribution of composite OLR anomaly in MJO [(a) active 5678 (b) active1234 (c) inactive 5678 (d) inactive1234] phases during the identified HW events over Indian region. The units of the shading considered is W/m^2 . The areas marked with dot signify a 90% level of significance.

in turn the impact on HW dynamics in India. During HW events the MJO activity in wet phases 1, 2, 3 and 4 have been associated with significant strong negative OLR anomaly in the tropical Indian ocean and significant strong positive OLR anomaly over northern parts of Indian land-mass (Fig. 7 b).

The composite anomaly of OLR during identified HW events in dry phases 5, 6, 7 and 8 is positive over the whole India except the northern Bay of Bengal region (Fig. 7 a-d). During the active wet phase, the OLR anomaly is negative which leads to the poleward movement of upper-level high pressure ridge (Fig. 7 b). Significant composite OLR anomalies have been identified over the Indian Ocean and India in all the MJO phase clusters, except for the central parts of India during the dry phase. This shows the enhanced convective activity over Indian ocean in the wet phase which leads to strengthen the high-pressure systems through the modulated tropical easterly jet streams. The composite anomaly of GPH and wind vector during the HW events at 500 hPa has been analyzed to understand the atmospheric pressure system and circulation. The significantly anomalous high-pressure system over northern India is related to sub-tropical high and associated anomalous anticyclonic flow of wind represents the blockage of high-pressure system. In the MJO active dry phase, cyclonic low-pressure system in Arabian sea and Bay of Bengal along with the elevated 500 hPa level over northern parts of India and Pakistan (Fig. 8a). The anticyclonic motion of wind over NW region imports and blocks hot air in eastern parts of country. The composite GPH increased during the inactive (suppressed convection) and decreased during the active (enhanced convection) (Fig. 8 a-d). During the wet phase, three anticyclonic high-pressure systems has been observed (Fig. 8b, d). There is a symmetric pattern of high-pressure system over NW region centered by high(low) pressure system in wet (dry) phase. The positive (negative) anomalies of GPH over north China, north Pacific Ocean has been observed in wet (dry) phase of MJO, which

is linked to presence of quasi stationary Rossby wave. The composite anomaly of VP during the identified HW events at upper 200 hPa and lower 850 hPa has been analyzed and shown in Figs. 9 and 10 in four clusters of MJO phases. At both the tropospheric levels, intensity of anomalous VP is significantly reduced with the weaker MJO and the reversed pattern of convergence at 200 hPa with divergence at 850 hPa has been observed over India. In the inactive wet phase, the least positive (negative) VP anomalies has been observed at lower (upper) tropospheric level (Fig. 9d,10d) which represents the blockage of air mass during the HW days. The converse effect has been observed in active dry phase as inferred from Figs. 9a and 10a. The moderate anomalies of convergence (divergence) at upper (lower) tropospheric level in MJO active wet (inactive dry) has been observed, which has a direct impact on the regional HW occurrence probability as shown in Figs. 9b-c and 10b-c over different parts in the country. The significant maximum positively anomalous SST has been observed in the inactive wet phase of MJO (Fig. 11 d). The significant higher SST during the wet phases has been observed over the regions as compared to the dry phase (Fig. 11 a-d) as the MJO active phase is known for deep convection.

4. Discussion

The climate is hot and dry during pre-monsoon summer season (April–June) accompanied by HW and mango showers in India. HW over India are triggered by high pressure systems over NW and Tibetan plateau region. This analysis aims to identify the regional variations in HW characteristics associated with the specific MJO phases. The new finding of this study is the spatial estimation of HW intensity and maximum duration, frequency in different phases of MJO. The intensity of HW over NW region is mostly affected by the strength of MJO, weaker MJO leads to reduction in intensity by 0.5 °C over the central region of

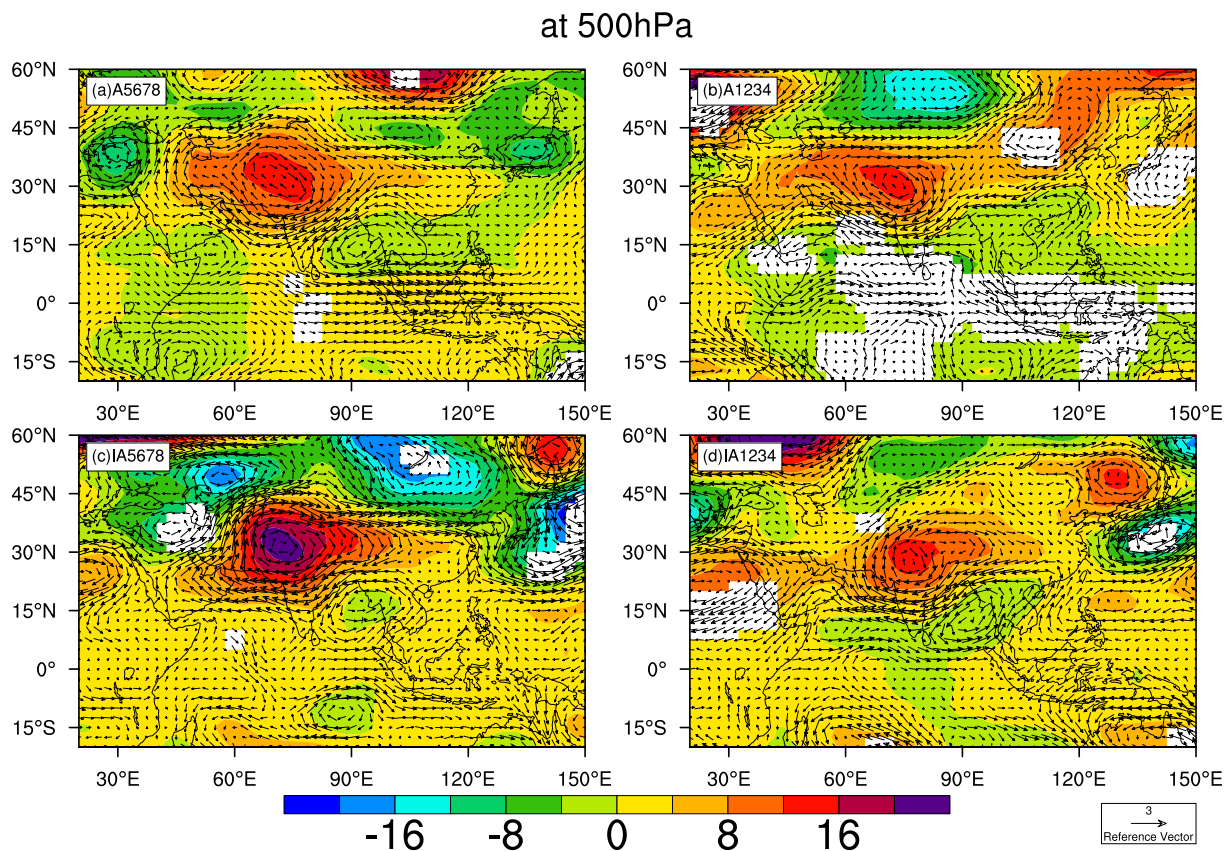


Fig. 8. Spatial distribution of composite GPH anomaly (shaded) overlaid with resultant wind anomaly vector at 500 hPa during MJO (a) active 5678 (b) active 1234 (c) inactive 5678 (d) inactive 1234 phases during the identified HW events over Indian region. The units of the shading considered is m/day. The areas shaded indicate a significance level of 90%.

at 200hPa

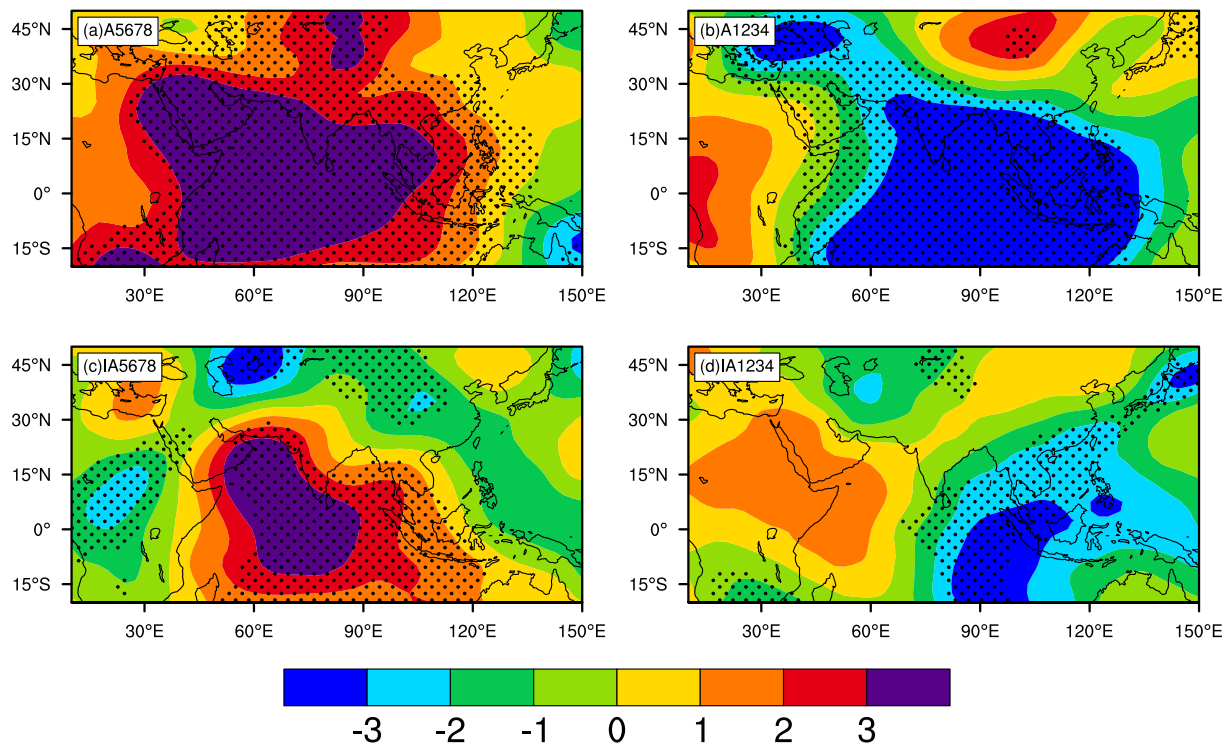


Fig. 9. Spatial distribution of VP composite anomaly at 200 hPa in MJO (a) active 5678 (b) active1234 (c) inactive 5678 (d) inactive1234 phases during the identified HW events over Indian region. The units of the shading are expressed as $10^5 \text{ m}^2 \text{ s}^{-1}$. The areas marked with dot signify a 90% level of significance.

at 850hPa

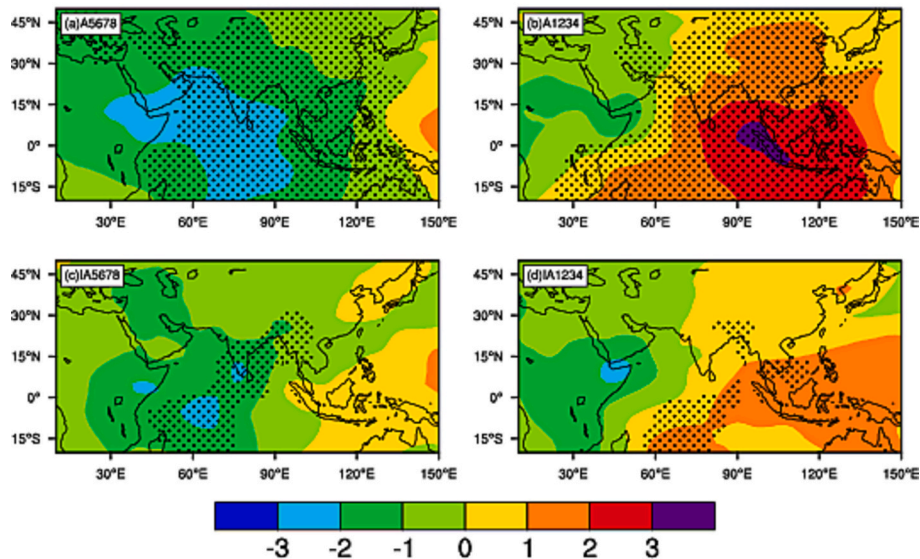


Fig. 10. Spatial distribution of VP composite anomaly at 850 hPa in MJO [(a) active 5678 (b) active1234 (c) inactive 5678 (d) inactive1234] phases during the identified HW events over Indian region. The units of the shading are $10^5 \text{ m}^2 \text{ s}^{-1}$. The areas marked with dot signify a 90% level of significance.

India. The eastern states HW intensity is approximately 5°C – 6°C in the active wet, active dry and inactive wet phase. The HW maximum duration decreases by 2 days on average over northern MP, NW and north central parts of India in the MJO active wet and dry phase than inactive wet and dry phase (Fig. 5). The probability of HW occurrence found to be maximum in the MJO phase 2–3 (Fig. 1) as the MJO

convection center lies over Indian Ocean. There is 2-fold rise in HW occurrence probability in the inactive MJO dry phase (5–8) and wet phase (1–4) has been observed mainly over SE coastal and NW region in India (Fig. 3). This contrast of change in the HW occurrence probability has been observed between SE, eastern and NW, north central, NE parts of India also marked by earlier study by Zore et al. (2021). The SE and

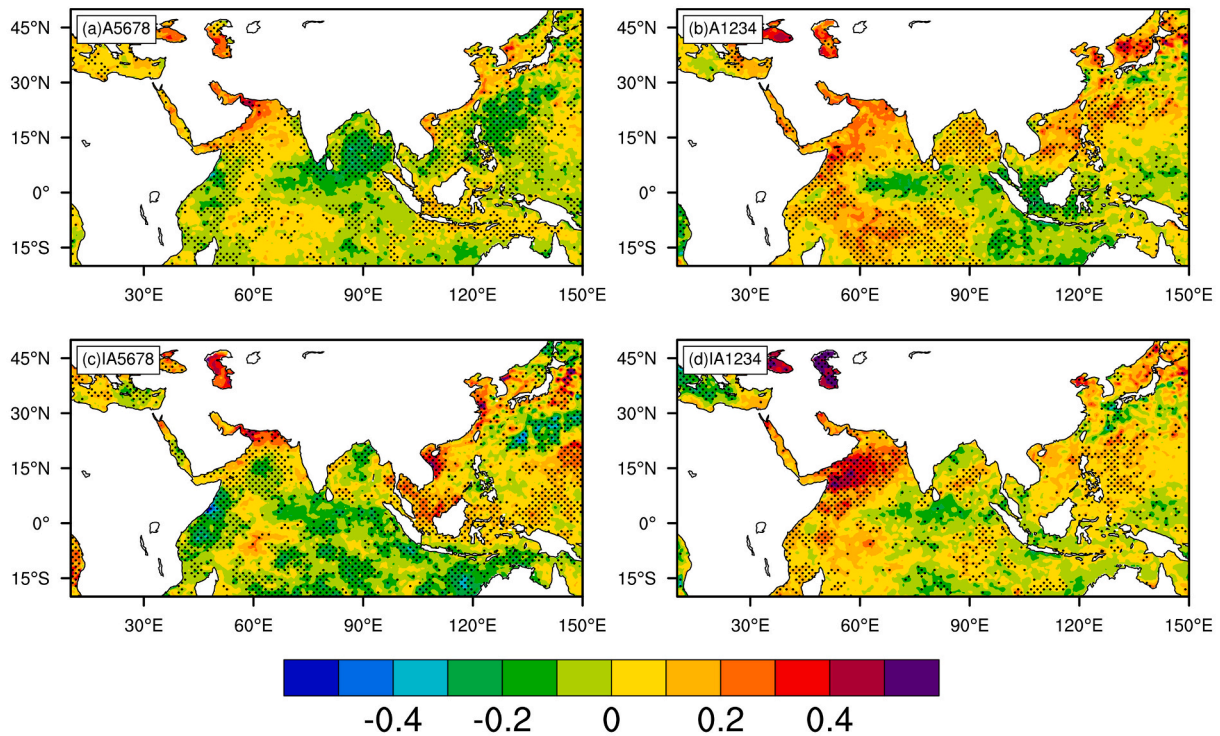


Fig. 11. Spatial distribution of SST composite anomaly ($^{\circ}\text{C}/\text{day}$) in MJO [(a) active 5678 (b) active 1234 (c) inactive 5678 (d) inactive 1234] phases during the identified HW events over Indian region. The areas marked with dot signify a 90% level of significance.

NW contrast can be linked with cyclonic low-pressure system in Arabian sea and Bay of Bengal in MJO active dry phase (Fig. 8a). The more likelihood of HW occurrence over NW, north central and NE states in the wet phase has been dominated with the presence of Rossby wave train (Fig. 8 b, d). The same has been evidenced, as MJO-induced Rossby wave trains play a crucial role in mediating the MJO's impact on weather and climate in the mid-latitudes by transmitting atmospheric perturbations and altering the typical atmospheric circulation patterns in those regions (Song and Wu, 2019). Also, the eastward moving enhanced convection located over Indian ocean in the wet phase is releasing the latent heat causes the upper-level high pressure ridge to shift poleward. The same is also shown in Fig. 7 (b, d), negative OLR anomaly in active wet phase is extending into southern India and in the inactive wet phase, is observed up to Indian ocean only. As a result of this MJO can modulate the upper-level circulation and formation of high-pressure systems over India. In the active dry phase, the elevated 500 hPa level over the India explains the increased chances of HW occurrence over the eastern part particularly over AP, Jharkhand, Chhattisgarh and Odisha as the cyclonic motion helps in creating the westerlies anomaly which further weaken the land sea breeze (Fig. 8a). The more likelihood of HW over eastern and SE coastal region observed (Fig. 3) can be linked with the high positive (negative) anomalous SST in the south China sea and northern Arabian ocean (Bay of Bengal) accompanied by the anticyclonic (cyclonic) movement of air at 500 hPa (Fig. 11a, 8a). The new finding of this study is the evolution of the dynamical linkage in MJO dry phase, SST and HW in India. Similarly, in wet phase, enhanced longer duration HW occurrence with low intensity temperature over NW, western coastal and central India noted to be linked with the high positive SST anomalies over western Arabian sea, Caspian Sea and Black Sea (Fig. 11b,d). The maximum positively anomalous SST has been observed in the inactive wet phase of MJO and same has been validated by observed anticyclonic high pressure wave pattern as presented in the Fig. 8d. The subsidence of air also suppresses the sea breeze and supports the HW formation over the coastal regions (Lee and Grotjahn 2019). Positive VP anomalies along with OLR can also

favor the formation and extension of HW (Chen et al., 2020). The least positive (negative) VP anomalies has been observed in the inactive wet phase at lower (upper) tropospheric level as evident from Figs. 9d and 10d, which represents the blockage of air mass during the HW days and the opposite effect has been observed in active dry phase as inferred from Figs. 9a and 10a. The moderate anomalies of convergence (divergence) at upper (lower) tropospheric level in MJO active wet (inactive dry) has been observed, which has a direct impact on the regional HW occurrence probability as shown in Figs. 9b-c and 10b-c over different parts in the country. To identify the localized effect of MJO on HW duration, MJO events in the lag correlation with HW is needed. The impact of MJO phases 5, 6, 7, and 8 needs to be analyzed as the convection center lies over the Pacific Ocean where SST anomalies plays an important role in HW over India. There is a need to quantify the combined effect of ENSO and MJO on HW as during the wet phases, the higher SST is observed. Also, the higher SST in the Indian ocean have been linked with the positive phase of ENSO.

5. Conclusions

The MJO stands as a prominent driver of tropical intra-seasonal variability, exerting influence on worldwide weather patterns. It is characterized by the eastward propagation of large-scale atmospheric convection and upper circulation anomalies. The innovation of this study is the modulation of spatial characteristics (in terms of the frequency, intensity and duration) of HW over India by the strength and 8 phases of MJO using RMM index data.

- The maximum likelihood of HW occurrence over most of the parts of India has been noted in MJO phase 2 and 3.
- The climatological probability of HW occurrence over India rise by >1-fold in the MJO inactive phase than the active phase over corresponding location in the wet phase. Overall, significantly, the wet (dry) phase leads to the doubling (total suppression) of HW occurrence probability over most of the parts of India.

- The HW maximum duration decreases by 2 days on average over northern MP, NW and north central parts of India in the MJO active than inactive phase.
- The maximum average duration (5–9) days has been observed in inactive dry and active wet phase.
- The HW intensity found to be reduced by 0.5 °C in the active wet phase over NW parts than inactive wet phase whereas during active dry phase increment of 0.5 °C have been noticed over northern MP and Vidarbha region.
- The differential dynamics of change in HW occurrence probability over eastern, SE and NW India are also supported by the elevated GPH accompanied by anticyclonic (cyclonic) wind circulation over Indian landmass (Indian ocean) in the wet (dry) phase. The elevated GPH in Arabian sea, northern India and south China sea representing the presence of quasi stationary Rossby waves train which mediates the impact of MJO on HW. Analysis of VP indicates that the subsidence of warm air into India from the upper to lower tropospheric levels in MJO phases during HW also supports the intensification of HW in inactive wet phase.

By examining the relationship between MJO phases and HW, this study can provide insights into the potential impacts on sectors like crop yields, water availability, energy demand, summer disasters, and overall economic productivity of the country. HW pose serious health risks, particularly for vulnerable populations. By understanding the influence of dynamical MJO phases on the regional HW disasters, public health authorities can develop targeted interventions and strategies to protect human health and reduce heat-related morbidity and mortality. Understanding the factors that contribute to the occurrence and intensity of HW is crucial for effective preparedness, mitigation, and adaptation strategies and this study will surely act as a guidance for the HW disaster management. The findings will contribute to a better understanding of the role of large-scale dynamical processes in modulating the region wise HW characteristics, aiding in the development of improved forecast methodology using the numerical weather prediction models for the accurate and advanced extreme HW events and mitigation strategies.

Credit author statement

Rani Devi and K C Gouda designed this study. All authors contributed to the data analysis and manuscript writing. All authors also contributed to the discussion and interpretation of the manuscript. All of the authors reviewed the manuscript. **K C Gouda** supervised the work.

CRediT authorship contribution statement

Rani Devi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **K.C. Gouda:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Smrutishree Lenka:** Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We acknowledge IMD, NOAA, NCEP and USGS for the data support. The authors RD and SL acknowledge UGC and CSIR respectively for the fellowship support. We acknowledge the Department of Science and Technology and Ministry of Environment, Forest and Climate Change, Govt. of India, for the project support. Also thankful to CSIR 4PI for all support.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosres.2024.107250>.

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